GUIDO GAGLIARDI

PERSONAL INFORMATION

Guido Gagliardi is a PhD student pursuing a double degree in International PhD. Program in Smart Computing at the University of Pisa, Florence, and Siena (Italy) and in the Doctoral Programme in Engineering Science at KU Leuven (Belgium).

His research is focused on new explainable artificial intelligence techniques for the analysis of physiological signals to support the decision-making process in clinical scenarios. His idea is that the use of artificial intelligence approaches such as neural networks represents a great opportunity to deepen human knowledge about the functioning of the brain and, hence, improve human's overall quality of life.

My dream is to follow up with a research career, in which I will use eXplainable Artificial Intelligence (XAI) to investigate the human brain through neural networks, in clinical scenarios.

Name:	**GUIDO GAGLIARDI**
Domicile:	**Paus Adriaan VI-College Hogeschoolplein 3 - 4 3000 Leuven (Belgium)**
Tel:	**+39 329 64 68 654**
E-mail:	guido.gagliardi@phd.unipi.it guido.gagliardi@kuleuven.be
Date of birth:	**25 September 1996**
Birthplace:	**Pisa (PI)**
Fiscal Code:	**GGLGDU96P25G702B**
ORCID:	**0000-0003-2020-6439**

EDUCATION

NOV. 2021—PRESENT	TITLE	DOUBLE DEGREE OF DOCTOR OF PHILOSOPHY (PHD.)
	Course	**Smart Computing Doctoral Programme in Engineering Science**
	Institute	**University of Pisa, Florence, and Siena (Italy) KU Leuven (Belgium)**
	Expected Graduation Date	**Oct. 2024**

JOURNAL PAPERS PUBLICATIONS

Title	**Concept-wise granular computing for explainable artificial intelligence**
Contributors	Alfeo, A.L.; Cimino, M.G.C.A.; Gagliardi, G.
DOI	*https://doi.org/10.1007/s41066-022-00357-8*
Journal	Granular Computing
Abstract	Artificial neural networks offer great classification performances, but their internal model works as a black box. This can prevent their outcomes to be employed in real-world decision-making processes, e.g., in smart manufacturing. To address this issue, the neural network should provide human-comprehensible explanations for their outcomes. This can be achieved by exploiting domain concepts and measuring their importance for the classification. To this aim, we implement an information granulation process via a neural network specifically trained to represent data instances featuring the same (different) concept's item close to (far away from) each other. By combining the representations for each concept, we obtain the so-called conceptual space embedding. The classification is obtained by processing it via a neural network classifier. The conceptual space embedding (i) organises the

data instances according to their concepts-wise proximity, resulting in a very informative data representation; this translates into greater classification accuracy if compared to a concept-wise approach from the state-of-the-art; and (ii) encodes each concept in one of its parts; this enables the measurement of the importance of one concept by manipulating the corresponding part of the conceptual space embedding. The proposed approach has been tested with real-world data from smart manufacturing.

Title	**Improving emotion recognition systems by exploiting the spatial information of EEG sensors**
Contributors	*Guido Gagliardi; Antonio Luca Alfeo; Vincenzo Catrambone; Diego Candia-Rivera; Mario G. C. A. Cimino; Gaetano Valenza*
DOI	*https://doi.org/10.1109/ACCESS.2023.3268233*
Journal	**IEEE Access**
Abstract	Electroencephalography (EEG)-based emotion recognition is gaining increasing importance due to its potential applications in various scientific fields, ranging from psychophysiology to neuromarketing. A number of approaches have been proposed that use machine learning (ML) technology to achieve high recognition performance, which relies on engineering features from brain activity dynamics. Since ML performance can be improved by utilising 2D feature representation that exploits the spatial relationships among the features, here we propose a novel input representation that involves re-arranging EEG features as an image that reflects the top view of the subject's scalp. This approach enables emotion recognition through image-based ML methods such as pre-trained deep neural networks or "trained-from-scratch" convolutional neural networks. We have employed both of these techniques in our study to demonstrate the effectiveness of our proposed input representation. We also compare the recognition performance of these methods against state-of-the-art tabular data analysis approaches, which do not utilise the spatial relationships between the sensors. We test our proposed approach using two publicly available benchmark datasets for EEG-based emotion recognition tasks, namely DEAP and MAHNOB-HCI. Our results show that the "trained-from-scratch" convolutional neural network outperforms the best approaches in the literature, achieving 97.8% and 98.3% accuracy in valence and arousal classification on MAHNOB-HCI, and 91% and 90.4% on DEAP, respectively.

CONFERENCE PAPERS PUBLICATIONS

Title	**Fine-grain Emotion Recognition using Brain-Heart Interplay measurements and eXplainable Convolutional Neural Networks**
DOI	*https://doi.org/10.1109/NER52421.2023.10123758*
Contributors	*Guido Gagliardi; Antonio Luca Alfeo; Vincenzo Catrambone; Diego Candia-Rivera; Mario G. C. A. Cimino; Maarten De Vos; Gaetano Valenza*
Conference	**IEEE N.E.R. 2023**
Abstract	Emotion recognition from electro-physiological signals is an important research topic in multiple scientific domains. While a multimodal input may lead to additional information that increases emotion recognition performance, an optimal processing pipeline for such a vectorial input is yet undefined. Moreover, the algorithm performance often compromises between the ability to generalise over an emotional dimension and the explainability associated with its recognition accuracy. This study proposes a novel explainable artificial intelligence architecture for a 9-level valence recognition from electroen- cephalographic (EEG) and electrocardiographic (ECG) signals. Synchronous EEG-ECG information are

combined to derive vectorial brain-heart interplay features, which are rearranged in a sparse matrix (image) and then classified through an explainable convolutional neural network. The proposed architecture is tested on the publicly available MAHNOB dataset also against the use of vectorial EEG input. Results, also expressed in terms of confusion matrices, outperform the current state of the art especially in terms of recognition accuracy. In conclusion, we demonstrate the effectiveness of the proposed approach embedding multimodal brain-heart dynamics in an explainable fashion.

Title	**Automatic feature extraction for bearings' degradation assessment using minimally pre-processed time series and multi-modal feature learning**
Contributors	Alfeo, A.L.; Cimino, M.G.C.A.; Gagliardi, G.
DOI	https://doi.org/10.5220/0011548000003329
Conference	**Proceedings of the 3rd International Conference on Innovative Intelligent Industrial Production and Logistics (IN4PL 2022)**

Abstract

Emotion recognition from electro-physiological signals is an important research topic in multiple scientific domains. While a multimodal input may lead to additional information that increases emotion recognition performance, an optimal processing pipeline for such a vectorial input is yet undefined. Moreover, the algorithm performance often compromises between the ability to generalise over an emotional dimension and the explainability associated with its recognition accuracy.

This study proposes a novel explainable artificial intelligence architecture for a 9-level valence recognition from electroencephalographic (EEG) and electrocardiographic (ECG) signals. Synchronous EEG-ECG information are combined to derive vectorial brain-heart interplay features, which are rearranged in a sparse matrix (image) and then classified through an explainable convolutional neural network. The proposed architecture is tested on the publicly available MAHNOB dataset also against the use of vectorial EEG input. Results, also expressed in terms of confusion matrices, outperform the current state of the art especially in terms of recognition accuracy. In conclusion, we demonstrate the effectiveness of the proposed approach embedding multimodal brain-heart dynamics in an explainable fashion.

Title	**Improving an Ensemble of Neural Networks via a Novel Multi-class Decomposition Schema**
Contributors	Alfeo, A.L.; Cimino, M.G.C.A.; Gagliardi, G.
DOI	https://doi.org/10.5220/0011547900003332
Conference	**Proceedings of the 14th International Joint Conference on Computational Intelligence - Volume 1: NCTA**

Abstract

The need for high recognition performance demands increasingly complex machine learning (ML) architectures, which might be extremely computationally burdensome to be implemented in the real-world. This issue can be addressed by using an ensemble learning model to decompose the multi-class classification problem into many simpler binary classification problems, e.g. each binary classification problem can be handled via a simple multi-layer perceptron (MLP). The so-called one-versus-one (OVO) is a widely used multi-class decomposition schema in which each classifier is trained to distinguish between two classes. However, with an OVO schema each MLP is non-competent to classify instances of classes that have not been used to train it. This results in classification noise that may degrade the performance of the whole ensemble, especially when the number of classes grows. The proposed architecture employs a weighting mechanism to minimise the contribution of the non-competent MLPs and combine their outcomes to effectively solve the multi-class classification problem. In this

work, the robustness to the classification noise introduced by non-competent MLPs is measured to assess in what conditions this translates in better classification accuracy. We test the proposed approach with five different benchmark data sets, outperforming both the baseline and one state-of-the-art approach in multi-class decomposition algorithms.

SEP. 2019—SEP. 2021	TITLE	MASTER'S DEGREE
	Course	**Artificial Intelligence and Data Engineering**
	Institute	**University of Pisa**
	Graduation Mark	**110 *cum Laude***
	Graduation Date	**24/09/2021**
	Exams' mark weighted average	**30.0**

Thesis Title — **A novel multimodal feature learning architecture for explainable affective computing**

Thesis Supervisors — **Mario G.C.A. Cimino, Gigliola Vaglini, Antonio Luca Alfeo**

Thesis Abstract — Development of AI architectures apt to recognize emotional states from electroencephalogram and other physiological signals and to provide insights on how the output is established.

MASTER'S DEGREE EXAMS

Code	Title	Mark/30
878II	DATA MINING AND MACHINE LEARNING	27
883II	LARGE-SCALE AND MULTI-STRUCTURED DATABASES	30
876II	CLOUD COMPUTING	30
875II	BUSINESS AND PROJECT MANAGEMENT	30
696AA	OPTIMIZATION METHODS AND GAME THEORY	27
891II	ROBOTICA E MACCHINE INTELLIGENTI	**30 cum Laude**
882II	INTERNET OF THINGS	30
886II	MULTIMEDIA INFORMATION RETRIEVAL AND COMPUTER VISION	30
888II	PROCESS MINING AND INTELLIGENCE	**30 cum Laude**
910II	INDUSTRIAL APPLICATIONS	30
877II	COMPUTATIONAL INTELLIGENCE AND DEEP LEARNING	30
893II	SYMBOLIC AND EVOLUTIONARY ARTIFICIAL INTELLIGENCE	**30 cum Laude**
	AVERAGE SCORE	30.0

SEP 2015—FEB 2019	TITLE	BACHELOR'S DEGREE
	Course	**Computer Engineering**
	Institute	**University of Pisa**
	Graduation Mark	**105/110**

Thesis Title: — **Novelty data detection methods**

Thesis Supervisors — **Francesco Marcelloni, Alessio Bechini**

Thesis Abstract — AI deployment on outlier detection task, based on two algorithms of LE (learning entropy) and ELBND (error and learning-based novelty detection).

System testing with both general stream of data and in presence of concept drift.

SEP 2010—JULY 2015	TITLE	HIGH SCHOOL DIPLOMA

Institute	Liceo Scientifico Ulisse Dini
Mark	87/100

RESEARCH PROJECT CONTRIBUTIONS

Title **Heartbeat anomalies detection using a Data Mining approach**

Course Data Mining and Machine Learning

Abstract The project investigated the possibility to use classical Data Mining approaches such as Support Vector Machines, k-Nearest Neighbours or Random Forest classifiers to face a multi-label classification problem in a strong unbalanced scenario. Specifically, the task was to perform arrhythmia detection on 5 classes of arrhythmia on the MIT-BIH dataset.
The project work focused, in the first step, on signal analysis and signal intelligence, to extract as many as possible useful features from the data; then the work moved to data oversampling to face data unbalancing and finally, **we proposed the construction of an ensemble of classifiers** to obtain good performance relative to the metrics adopted.

Title **Medical image analysis using CNN**

Course Computational Intelligence and Deep Learning

Abstract The project investigated the possibility to use Convolutional Neural Networks to perform abnormalities detection in medical images such as mammography for cancer detection.

In the first instance, we constructed from scratch Convolutional Neural Networks to face up the problem, then we tried to build a Fine-Tuning model and a Siamese one to achieve higher performances.

Eventually, since the main problem on the task was that the data were strongly unbalanced, **we proposed and constructed a Generative Adversarial Network approach, in particular a Variational Autoencoder,** to oversample the dataset in a way that would let the Convolutional Neural Network train more without going in overfitting too fast due to the replication of images usually done when dealing with oversampling in images data.

Title **User mobility profile**

Course Industrial Application

Abstract The project investigated the possibility to use an Artificial Intelligence approach on Real Time-systems such as a new generation of unmanned cars, to create a fast and responsive user profiling, storing all the user information, and automatically associate this profile with a physical person whenever he/she enter in the car with an automatic face recognition step.

My contribution to the teamwork has focused on **the design of the Artificial Intelligence system, with Deep Neural Network and OpenCV, in a Real-Time and mobile system** with its typical constraints (low power consumption, low bandwidth, lossy networks and so on), both from the point of view of the **implementation of the AI** and the **design of a cloud platform** needed to maintain consistency in the data.

Title **A Deep Reinforcement Learning approach to train multi-agent robot system for formation control**

Course Robotica e Macchine Intelligenti

Abstract The project investigated a very unexplored research topic: the use of a Deep Reinforcement Learning approach to training a multi-agent robot network, in a decentralized fashion, to let the robots perform a leader-follower task with collision avoidance constraints.

The main contributions of the research project, on which I worked alone, have been: the study and the **production of a new environment's reward functions** able to fit both the leader-follower task and the Actor-Critic Deep Reinforcement Learning model; the **study of the Actor-Critic model in a continuous action space** (the robots were treated as integrators, so they were controlled in speed) and finally the construction of both **a custom simulation environment**, using Google open-ai gym, and a **new training framework** capable of letting the agents correctly visit the state space and follow the leader.

Title	**Interior Point Method improvements using sparse matrixes**
Course	Symbolic and evolutionary Artificial Intelligence
Abstract	The project investigated the possibility to improve the speed performances of the Interior Point Method algorithm in a multi-objective environment using sparse matrixes to solve in a faster way the Karush Kuhn Tucker system related to the method.

My contribution to the research has been done in the first moment in the extension of the actual Julia 1.6 libraries and the BAN (Bounded Algorithmic Numbers) library provided by the University of Pisa, to explicitly support the concept of sparse matrixes while using non-Archimedean numbers (fundamental in a multi-objective lexicographic optimization problem).

Then the study continued by proposing new analytical methods to solve the Karush Kuhn Tucker system, such as matrix partitioning and Cholesky factorizations, in a sparse matrix environment, to speed up the computation compared to the LU factorization needed to solve the standard system.

STUDY PLAN RELATED EXPERIENCE

I have participated in the **Roborace** team of the **Research Centre E. Piaggio of the University of Pisa**, in 2018, (ref. Engineer Danilo Caporale, Research Centre E. Piaggio) referenced by Prof. Lucia Pallottino of the University of Pisa, which aims to build an intelligent autonomous system to drive an electrical unmanned race car, for competing with other universities around Europe.

My work was focused on the localization of the vehicle and the map construction in absence of GPS signal only with lidar systems.

In particular, I had the opportunity to learn and understand the main problems and requirements that this kind of system needs to face while driving, and also, I needed to learn the ROS language (that I've further improved during my course of "Robotica e Macchine Intelligenti") for simulation purpose and hardware virtualization in a real-world scenario.

Furthermore, during this period I also **have been in the UK at the Roborace company headquarters**, on a mission for the University of Pisa**, to test the performances of the implementations of our team on the real race care prototype**.

SKILLS AND ABILITY

MOTHER TONGUE	ITALIAN
OTHER LANGUAGES	**ENGLISH**
Reading	C1
Writing	C1
Listening	C1
Speaking	C1

TECHNICAL SKILLS

Artificial Intelligence, Deep Learning, Data Mining, Machine Learning, Robotics, Real-Time Programming, Internet of Things, Networking.

C++, Python, Julia, ROS, Java, UML.

RELATIONAL SKILLS

Ability to work well in a team acquired both through academic group projects and by participating in University of Pisa Radio, RadioEco. This last activity also helped me acquire leadership and decision-making skills because of my position as President of the Association, which lasted 2 years, and also my public speaking capabilities.

Aptitude to work in well competitive and stressful scenarios with close deadlines as proven during my academic career.

EXTRACURRICULAR ACTIVITIES

From 2017 to 2021 I took part in RadioEco a non-profit student Association, which, in collaboration with the University of Pisa, provides a professional Radio-Station laboratory for students. RadioEco counts a big number of members, with at least 300 applications a year. The Association has an elective executive board that is in charge of all the activity that the Association propose for its members and the public, and it manages also all the funds and the technical equipment provided by the University of Pisa. The executive board is led by an elective president in charge for 2 years.

July 2019 – June 2021 Title **President of the Association**

Description Full legal and managerial responsibility of the Association, head of the executive border and station managing of the radio.

Sep 2018- July 2019 Title **IT-Manager in the executive board**

Description General managing of the Association, related to the IT infrastructure, such as the website and the radio flow streaming.

Sep 2017 – Dec 2020 Title **Radio Speaker**

Description Hosting and directing many radio shows, such as UnipiNews, with insights on social and cultural-related topics.

CERTIFICATES AND DOCUMENTS

Master Degree in Artificial Intelligence and Data Engineering

Bachelor’s Degree in Computer Engineering

High School Diploma

Driver License B

English Certification C1

Il sottoscritto GUIDO GAGLIARDI nato a PISA prov. PI il 25/09/1996 residente a PISA prov. PI Andrea Cesalpino 9, 56124, consapevole delle sanzioni penali, nel caso di dichiarazioni non veritiere, di formazione o uso atti falsi richiamate dall'art. 76 del D.P.R. 445 del 28 dicembre 2000, nonché della sanzione ulteriore prevista dall'art. 75 del citato D.P.R. 445 del 28 dicembre 2000, consistente nella decadenza dai benefici eventualmente conseguenti al provvedimento emanato sulla base della dichiarazione non veritiera. Autorizzo il trattamento dei miei dati personali presenti nel cv ai sensi del Decreto Legislativo 30 giugno 2003, n. 196 "Codice in materia di protezione dei dati personali" e del GDPR (Regolamento UE 2016/679).

UNIVERSITÀ DI PISA

CENTRO LINGUISTICO

Guido Gagliardi

born in Pisa on September 25, 1996

attended a Course of ***English for Research Publication and Presentation Purposes*** of 30 hours from January 25, 2022 to March 29, 2022 (attendance: 80%) with the following result:

Grade*: Very Good

(*) Scale: Pass, Good, Very good, Excellent

The Director

Prof. Silvia Bruti

Handwritten signature omitted pursuant to art. 3 co. 2 of DL 12 February 1993 n.39

The present **Certificate of ATTENDANCE and ACHIEVEMENT** is given to students who attend at least 80% of the course and certifies that the student has fulfilled the **C1 Level** objectives for **Writing** (academic writing skills related also to scientific manuscripts) and **Speaking Skills** (presenting and debating the PhD student's research project). For the above-mentioned skills, the overall final level is comparable to **C1** (**CEFR**).

CLI – Centro Linguistico dell'Università di Pisa - Organizzazione con Sistema di gestione qualità certificato UNI EN ISO 9001:2015

Sede: via Santa Maria 36 - 56126 PISA (I) C.F.: 80003670504 P.I.: 00286820501

Segreteria Didattica: tel. 0502215590, fax: 0502210663, e-mail: cli@cli.unipi.it

Segreteria Amministrativa: tel. 0502215921, fax: 0502210662, e-mail: amministrazione@cli.unipi.it